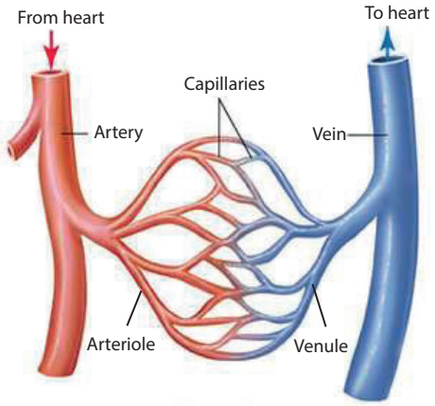


Figure 14.10 Structure of blood vessel

14.14 Types of Circulatory System

Animals possess two types of circulatory system. They are

1. Open type
2. Closed type

Open type

In open type the blood is pumped by heart into blood vessels that open into blood spaces called as **sinuses**. These sinuses are the body cavities which are called **haemocoel**. Capillary system is absent. e.g. Arthropods, Molluscs and Ascidians.

Closed type

In closed type the blood flows in a complete circuit around the body through specific blood vessels. The blood flows from arteries to veins through small blood vessels called capillaries. e.g Vertebrates.

More to Know

Closed circulatory system was discovered by William Harvey (1628) who is regarded the Father of Modern Physiology.

14.15 Structure of Human Heart

Heart is a muscular pumping organ that pumps out the blood into the blood vessels. Human heart is situated between the lungs, slightly tilted toward the left and above the diaphragm in the **thoracic cavity.** The heart is made of specialized type of muscle called the **cardiac muscle.**

The heart is enclosed in a double walled sac called **pericardium**. It contains lubricating **pericardial fluid** which reduces **friction** during heart beat and protects it from mechanical injuries.

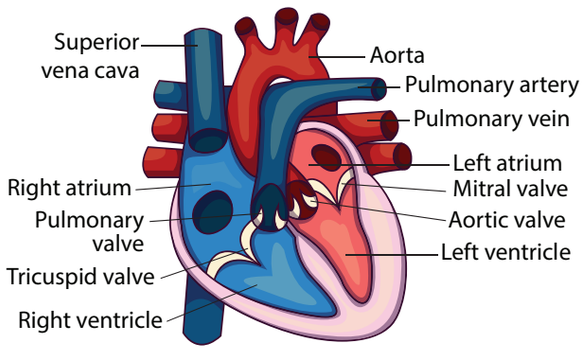


Figure 14.11 External structure of human heart

The human heart is four chambered. The two upper **thin** walled chambers of the heart are called auricle or **atria** (sing: atrium) and two lower **thick** walled chambers are called **ventricles.** The chambers are separated by partition called **septum.** The septum between auricles and ventricles prevents the mixing of oxygenated and deoxygenated blood.

The two auricles are separated from each other by **interatrial septum**.The left atrium is smaller than the right atrium. The right atrium receives deoxygenated blood from different parts of the body through the main veins **superior vena cava**, **inferior vena cava** and **coronary sinus. Pulmonary veins** bring **oxygenated blood** to the left atrium from the lungs. The right and left auricles pump blood into the right and left ventricles respectively.

The ventricles form the lower part of the heart. The two ventricles are separated from each other by an **interventricular septum**. The left and right ventricles have **thick walls** because the ventricles have to pump out blood with force away from the heart. From the **right ventricle**

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